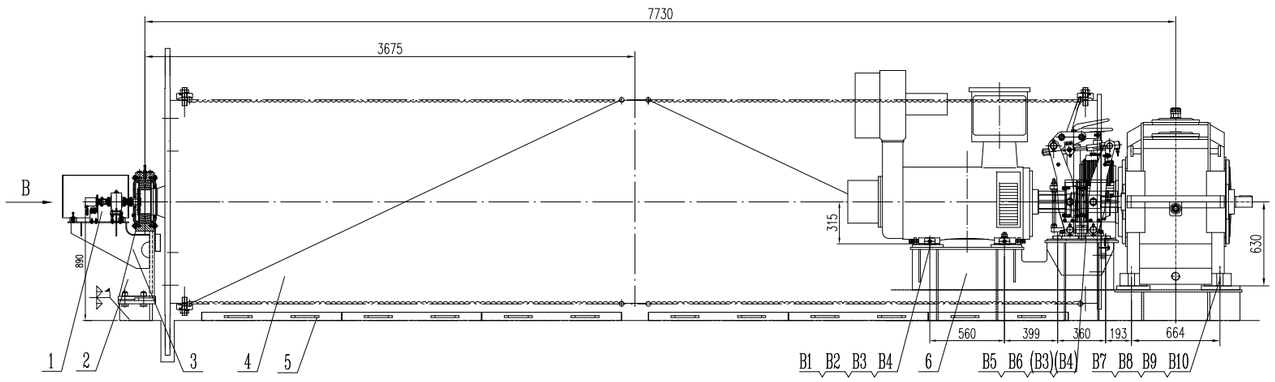
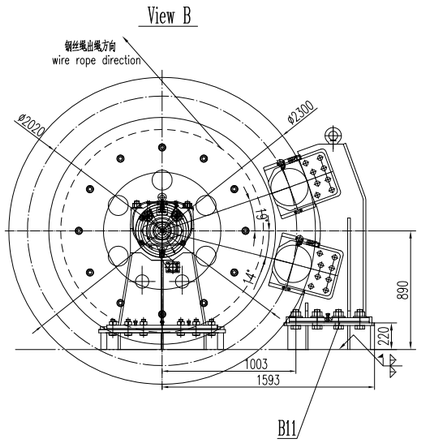
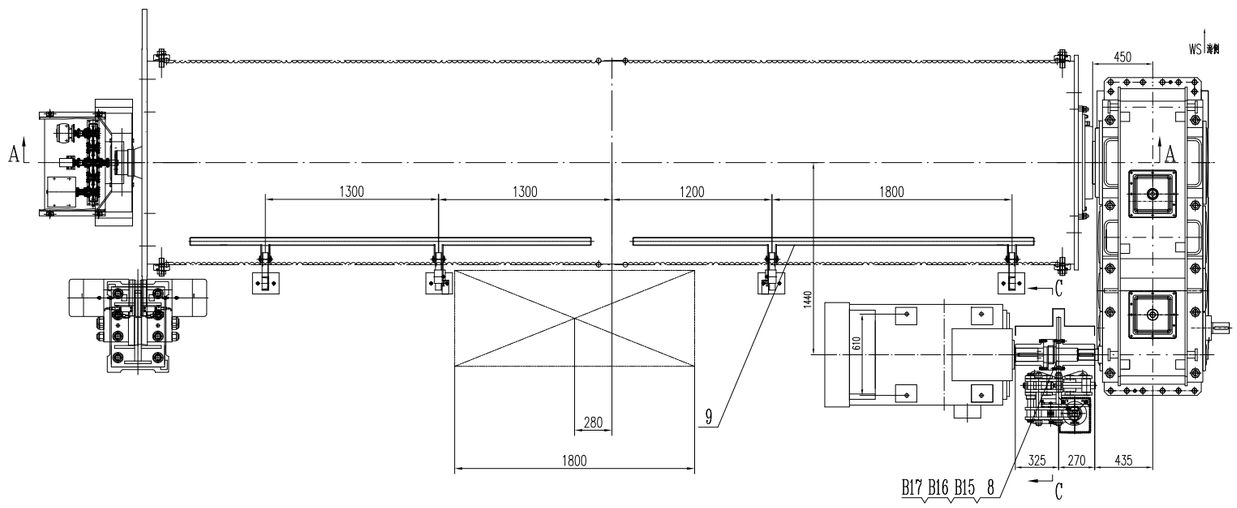
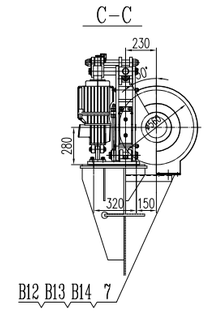
REV	DATE	BY	DESCRIPTION
01	2012.06.01	tonghui	
02	2012.08.17	tonghui	

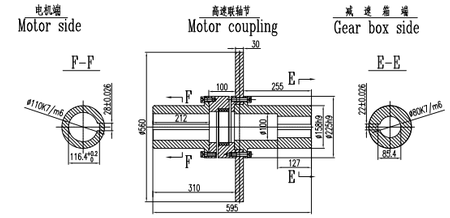




MAIN DATA 主要参数			
BOOM HOIST TIME (SINGLE) 大梁提升时间（单程）	0°–80°		≤5 min
MOTOR 电机 (YASKAWA)	TYPE	型号	BDK–IKMU
	RATED Duty,Power,Speed	额定参数	
	APPLIED POWER	功率	375kW
	APPLIED SPEED	转速	1750rpm
REDUCER 减速箱 (SEW)	TYPE	型号	ML 4PSF120X
	RATIO	速比	100.56
MOTOR COUPLING 高速联轴节 (KTR)	TYPE	型号	KTR AFN SB100–560x30
	RATED TORQUE	额定扭矩	4950 N.m
DRUM COULING 卷筒联轴节 (ZPMC)	TYPE	型号	DC10B
	RATED TORQUE	额定扭矩	560 KN.m
WIRE ROPE 钢丝绳 (TOYKO)	TYPE	型号	IWRC 6xP.WS(36)–ø35.5–B
	BREAKING STRENGTH	破断拉力	975 kN
MOTOR BRAKE 高速制动器 (ZPMC)	TYPE	型号	YP31–ED2000–560x30
	BRAKE TORQUE	额定制动力矩	5750 N.m
	SETTING TORQUE	设定制动力矩	5100 N.m
DRUM BRAKE 卷筒制动器 (ZPMC)	TYPE	型号	SB315–2300x36
	BRAKE TORQUE	额定制动力矩	2x254.52 kN.m
Drum 卷筒	Diameter	直径	ø1530 mm







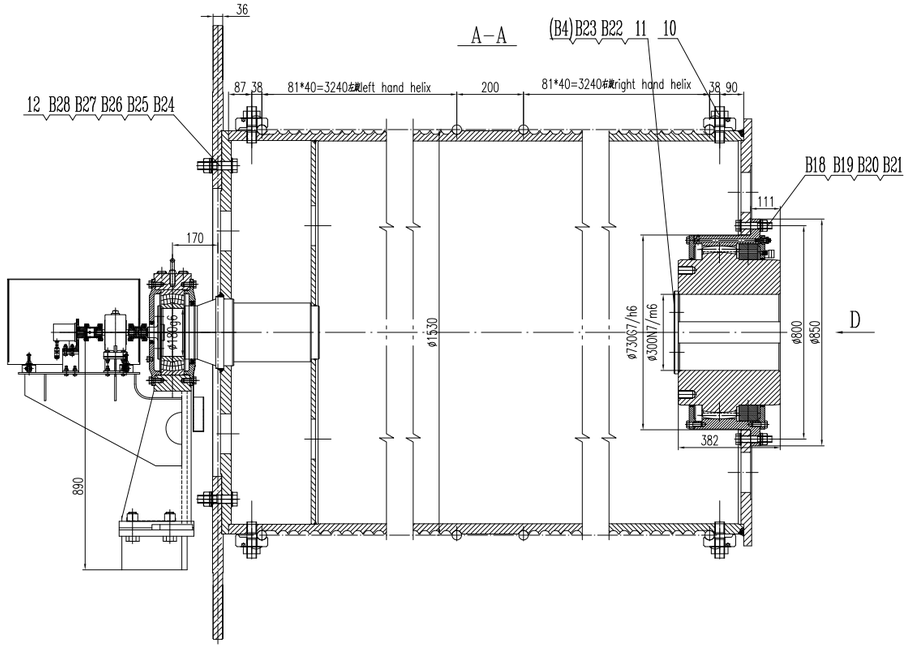
技术要求

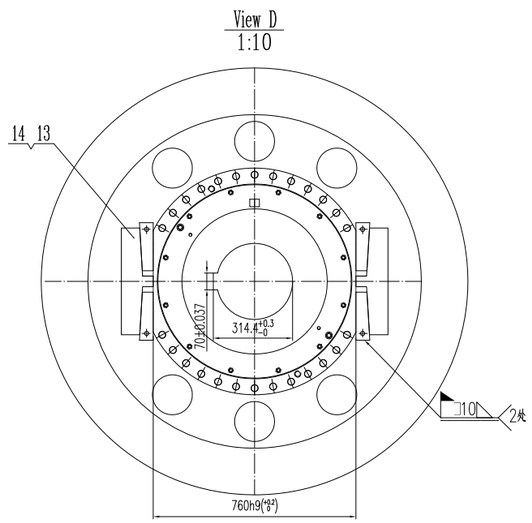
1. 电机中心线与减速箱中心线的交角(△b)应小于2'，径向相对偏差(△r)应小于0.05mm，见右图
2. 卷筒中心线与减速箱低速轴中心线的交角应小于10'
3. 装配时，轴承、齿形联轴器应加适量润滑脂。
4. 零部件装配完成后，定位采用楔形挡块，楔形挡块应有效固定；定位面有效接触，敲击力应不大于50kg，内侧定位块点焊。
5. 零部件装配完成后，定位采用螺栓挡块，螺栓挡块应有效固定；螺栓头与定位面有效接触，预紧力应不大于50kg，锁紧螺母拧紧。
6. 主要螺栓的预紧力见下表，卷筒制动盘、低速联轴节固定螺栓须在螺杆上均匀涂上二硫化钼润滑脂。
7. 联轴节、制动器等外购零部件安装时应遵守相关零部件使用说明书中的安装要求。
8. 卷筒制动盘、低速联轴节与卷筒螺栓法兰接触面装配前只允许涂环氧富锌底漆并须保持清洁、干燥、无油污等污染物。



Notes

1. The angular displacement (△b) between the axes of the high speed shaft of reducer and motor shall less than 2' and the radial misalignment (△r) shall less than 0.05mm. Please refer to figure right.
2. The angular displacement between the axes of drum and low speed shaft of reducer shall less than 10'.
3. Proper amount of grease shall be added into the bearings and gears of coupling.
4. The alignment taper shear blocks shall be well secured after machinery alignment: locating face contract each other sufficiently, the knocking force shall less than 50kg,inside taper blocks be tack welded.
5. The jacking screw shall be well secured after machinery alignment: The bolt of head and locating face contract each other sufficiently, the tighten force shall less than 50kg, lock nuts being tightened.
6. Tighten the bolts to the specified tension force as shown in below table. The bolts of drum coupling or drum brake disc shall be painted MoS_2 on the screw.
7. Installing instructions in the manuals of purchased parts such as coupling, brake and etc shall also be conformed.
8. Only epoxy zinc rich primer is permitted to be painted on the fray surface between drum and drum coupling or drum brake disc. The fray surface shall keep clean, dry, and non–polluted before assembly.





序号 Item No	B3	B9	B20二硫化钼	B24二硫化钼
螺栓尺寸D size	M24	M48	M24(10H)	M30(10H)
拧紧力矩Mo tightening torque	710 Nm	5900 Nm	735 Nm	1462 Nm
预紧力Fo tighten force	148.4 kN	618.8 kN	205.1 kN	326.2 kN
扭矩系数k torque coefficient	0.2	0.2	0.15	0.15

工号/JOB NO. ZP12–1873		项目名称/PROJECT Sri Lanka CICT 12 STS Cranes	
设计/DSGN	校对/CHKD	审核/EXAM	批准/APPD
日期/DATE	日期/DATE	日期/DATE	日期/DATE
材料/MAT	比例/SCALE 1:18	图名/DRAWING NAME 提升机构 Boom Hoist Device	图幅/FORMAT A1
重量/WEI (Kg) 24644			
ZPMC 上海振华重工集团		图号/DRAWING NO. J526A05	版次/REV 页次/PAGE

REVISION Description & Date

明 细 表
SUBSIDIARY LIST

*The quantity and weight of this list is for one set

*此表为一套的数量和重量

设计：

_21735_ kg/set, _1_ set/crane, _21735_ kg/crane

校对：

PROJECT NAME or CODE: Sri Lanka CICT 12 STS Cranes			SUB-PART NAME: Boom hoist device 俯仰机构			SUB DRAW NO: J526A05	
序号 NO.	图号或标准 SUB-NO.or STDD	名称 NAME	规格或型号 SIZE & STYLE	材料 MTRL	每套数量 Q'TY/SET	单重 PIECE	总重 TOTAL
1	J526A0501	限位装置 Limit switch		Ass.	1	124	124
2	J526A0502	卷筒轴承座 Bearing pedestel		Ass.	1	476	476
3	J526A0503	集油盘 Drip pan 1		Welded	1	0.9	0.9
4	J526A0504	卷筒 Drum		Welded	1	9253	9253
5	J526A0505	集油盘 Drip pan 2		Welded	6	20	120
6	J526A0506	底架 Base		Welded	1	395	395
7	J526A0507	联轴器罩壳 Coupling Cover		Welded	1	12.3	12.3
8	J526A0508	轴端挡板 1 Baffle 1		Q235	1	0.5	0.5
9	J526A0509	防跳槽装置 Peripheral guard		Ass.	2	74.7	149.4
10	RCA10	夹绳板压板组10 Clamp ass.10		Ass.	12	4.5	54
11	J526A0510	轴端挡板 2 Baffle 2		Q235	1	10.5	10.5
12	J526A0511	卷筒制动盘 Brake disk		A709-50-2	1	918	918
13	ZPDR10-1	楔形块 Block 1		35	2	1.45	2.9
14	ZPDR10-2	楔形块 Block 2		35	1	4.95	4.95
B1	BDK-IK-315KC2K	电动机 Motor	375kw,1750rpm	YASKAWA	1	1600	1600
B2	GB5782-86	螺栓 Bolt	M24x130	8.8s,Dacro.	4		
B3	ZN6175-T-SL	螺母 Nut	M24	8H,Dacro.	8		
B4	GB97.1-85	垫圈 Washer	24	200HV, Dacro.	10		
B5	YP31-ED2000	高速制动器 Service brake		ZPMC	1	236	236
B6	GB5782-86	螺栓 Bolt	M24x90	8.8s,Dacro.	4		
B7	ML 4PSF120X	俯仰减速箱 Gear box		SEW	1	6350	6350
B8	GB5782-86	螺栓 Bolt	M48x220	8.8s,Dacro.	10		
B9	ZN6175-T-SL	螺母 Nut	M48	8H,Dacro.	10		
B10	GB97.1-85	垫圈 Washer	48	300HV, Dacro.	10		
B11	SB315	低速制动器 Emergency brake	带底座订货	ZPMC	1	1880	1880
B12	GB5783-86	螺栓 Bolt	M8x30	不锈钢	4		
B13	ZN6175-T-SL	螺母 Nut	M8	不锈钢	4		
B14	GB97.1-85	垫圈 Washer	8	不锈钢	4		
B15	Rotex100 AFN-SB	高速级联轴节 Motor coupling		KTR	1	148	148
B16	GB5783-88	螺栓 Bolt	M20x50	8.8s,Dacro.	1		

明 细 表
SUBSIDIARY LIST

*The quantity and weight of this list is for one set

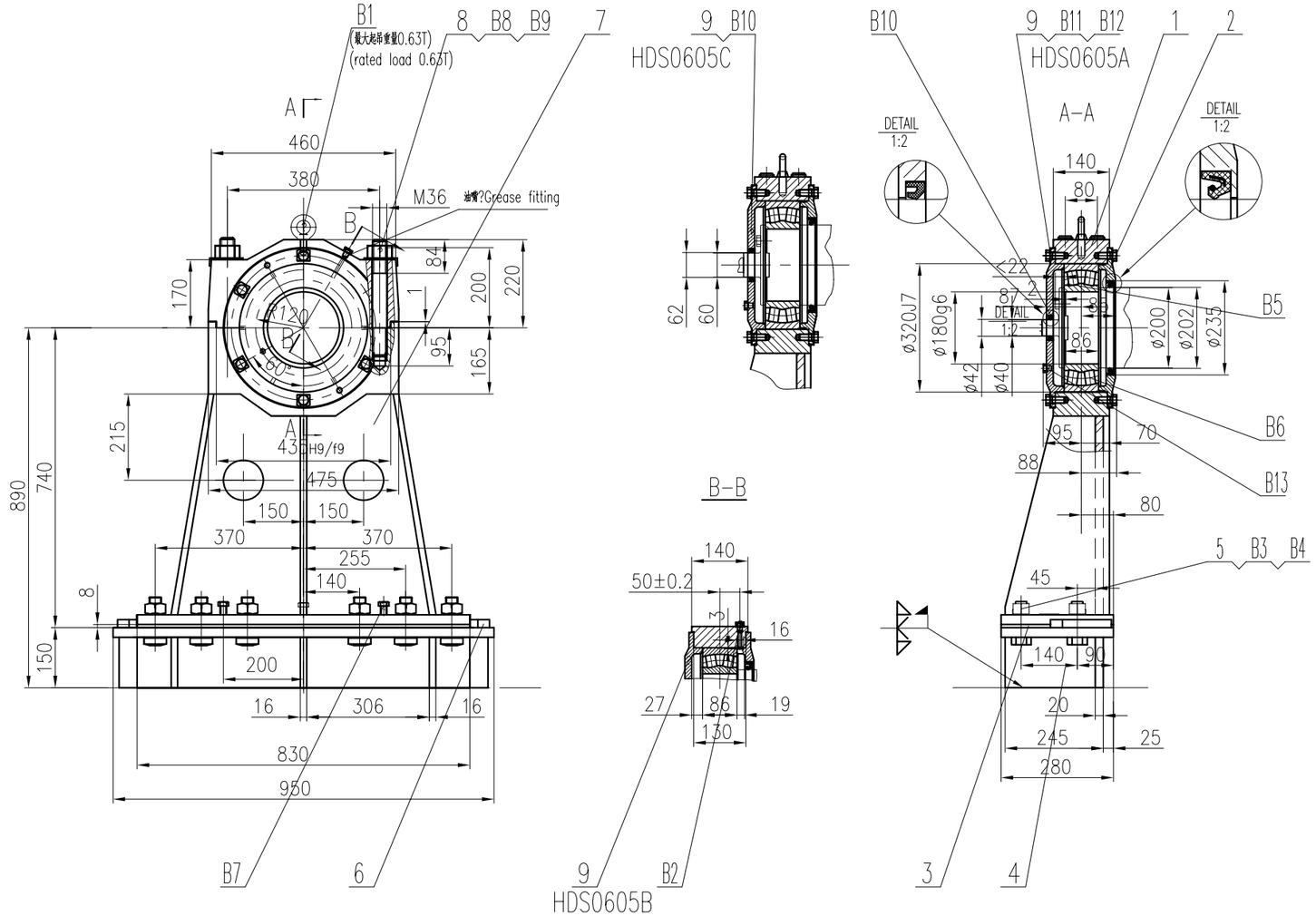
*此表为一套的数量和重量

设计：

_21735_ kg/set, _1_ set/crane, _21735_ kg/crane

校对：

PROJECT NAME or CODE: Sri Lanka CICT 12 STS Cranes			SUB-PART NAME: Boom hoist device 俯仰机构			SUB DRAW NO: J526A05	
序号 NO.	图号或标准 SUB-NO.or STDD	名称 NAME	规格或型号 SIZE & STYLE	材料 MTRL	每套数量 Q'TY/SET	单重 PIECE	总重 TOTAL
B17	GB856-88	垫圈 Washer	20	200HV, Dacro.	1		
B18	DC10B	卷筒联轴节 Drum coupling		ZPMC	1		
B19	GB5782-86	螺栓 Bolt	M24x120	10.9s,Dacro.	26		
B20	ZN6175-T-SL	螺母 Nut	M24	10H,Dacro.	26		
B21	GB97.1-85	垫圈 Washer	24	300HV, Dacro.	52		
B22	GB32.1-88	螺栓 Bolt	M24x60	8.8s,Dacro.	2		
B23		不锈钢丝	φ 2		1m		
B24	ZN6175-SL	螺母 Nut	M30	10H,Dacro.	14	其中两件发黑处理	
B25	GB97.1-86	垫圈 Washer	30	300HV, Dacro.	24		
B26	GB96.1-86	垫圈 Washer	30	300HV, 发黑	4		
B27	GB5782-86	螺栓 Bolt	M30x130	10.9s,Dacro.	12		
B28	GB27-88	螺栓 Bolt	M30x130	10.9s,发黑	2		



技术要求

1. 定位块ZP2现场烧焊.
2. 螺栓M16的预紧力矩为150Nm,M24为690Nm,M36为1800Nm.
3. 调整:通过调整件3"调节板"的厚度来调整中心高H,从而保证卷筒的另一端联轴节的对中性要求,调整垫片最多垫二层.
4. 加工轴承外园孔及端盖孔时,止口剖分面加1mm的工艺垫.
5. 装配时,端盖II上的注油孔布置在下部侧面,可作为放油孔.

Technical Requirement

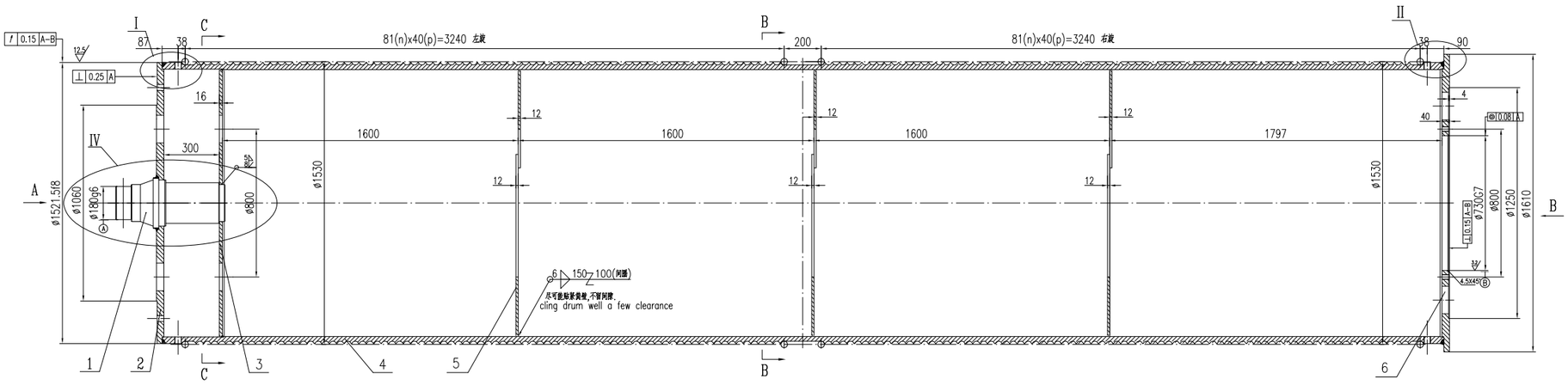
1. Piece 6 (ZP2) should be welded after . alignment at site.
2. Pretightening torque of bolt: M16 is 150Nm,M24 is 690Nm,M36 is 1800Nm.
3. Adjustment procedure:change height of H by using different thickness item 3 for alignment. Item 3 could not exceeded 2 piece.
4. Add 1mm washer on the split surface step of pedestal when machining bearing and hole of end cover.
5. Item B13 should be the place shown on the drawing,and it can be oil outlet

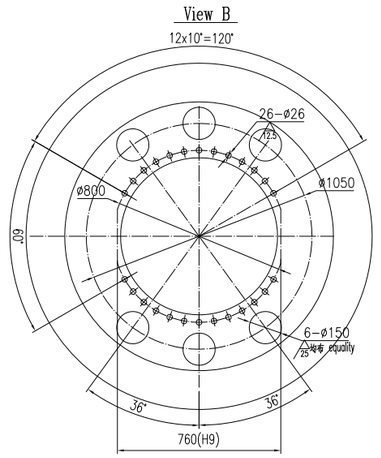
NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
B13	TP1	Bolt 螺塞 R1/4"	Q235	1			
B12	GB854-88	Washer 垫圈 16	Q235	12			
B11	GB5783-86	Bolt 螺栓 M16x35	8.8s,Dacro.	12			
B10	HG4-692-67	oil seal骨架油封PD40x62x12		1			
B9	GB854-88	Tab-washer 止动垫片 36	Q235	4	0.05	0.2	
B8	GB6170-86	Nut 螺母 M36	8H,Dacro.	4	0.6	2.4	
B7	GB5783-86	Bolt 螺栓 M16x45	8.8s,Dacro.	3	0.08	0.24	
B6	SKF/FAG	Bearing 轴承 22236 CA		1			ø180Xø320X86
B5	HG4-692-67	Oil seal 油封PD200x235x18		1			
B4	GB6170-86	Nut 螺母 M30	8H,Dacro.	12	0.18	2.21	
B3	GB5782-86	Bolt螺栓 M30x100	8.8s,Dacro.	12	0.72	8.67	
B2		Oil nozzle 油嘴 PT1/4"	黄铜 brass	1	0.06	0.12	黄铜 Brass
B1	GB825-88	Screw 吊环螺钉 M16		1	0.24	0.24	

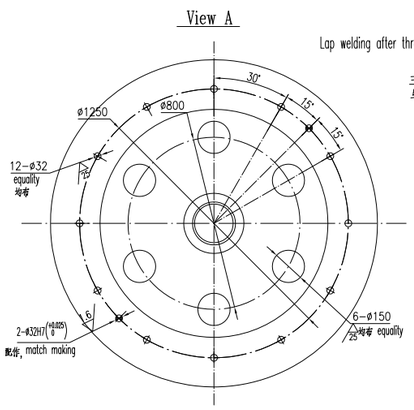
NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	QTY/set	each WEIGHT	total WEIGHT	NOTE
9	HDS0605A	end cover端盖II	Q235	1		17.8	
8	HDS0604(GB900-88)	stud 螺柱 M36x220	45,8.8s	4	1.94	7.76	热镀锌HOT-DIP.GAL
7	J526A050201	down pedestal下轴承座	welded	1	167.5	167.5	中心高选择select
6	ZP2	locating block 2号定位块	Q235	2group	1.1	2.2	
5	HDS0205	washer垫片	Q235	6	0.21	1.26	借用borrow
4	HDS0204	support底座	welded	1		111.2	借用borrow
3	HDS0203	adjusting plate调整垫	Q235	1		76.5	借用borrow
2	HDS0602	end cover端盖1	Q235	1		12.3	
1	HDS0601-1I	upper pedestal上轴承座	Q235	1		66	PT1/4"

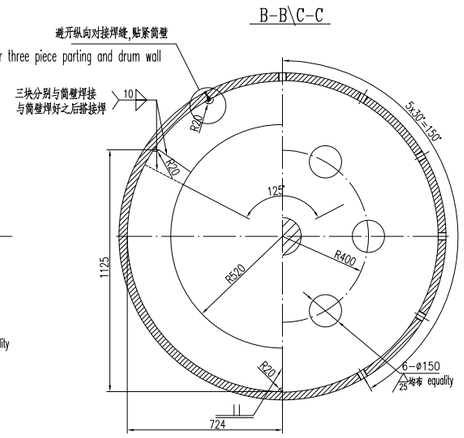
J526A0502	Bearing Pedestal 卷筒轴承座组件				476 kg	1:10

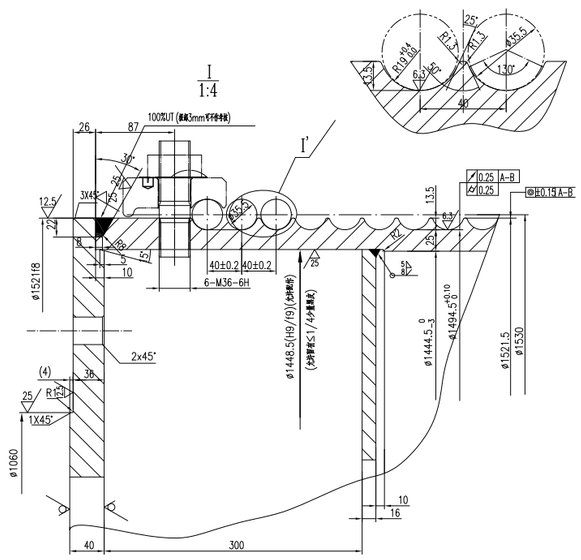
ZPMC						
DRAW			TECH.		TRACE	
CHCK			STDD		CHCK	

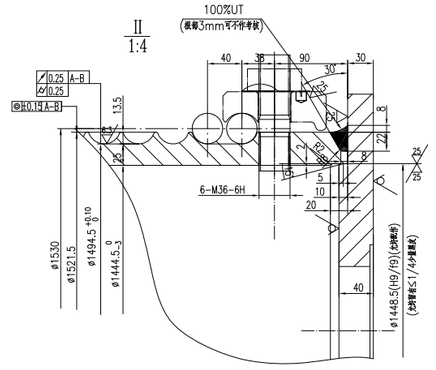


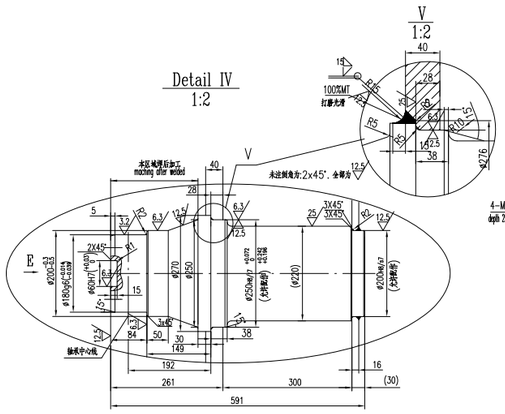


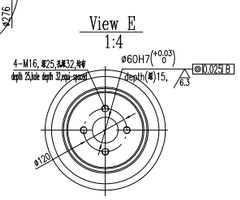


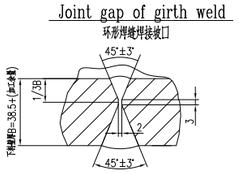


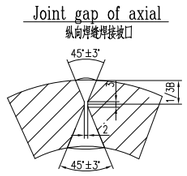


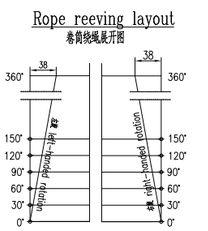












技术要求

1. 卷筒按如图所示.
2. 卷筒用Q345B钢板冷卷焊接制成.
3. 卷筒体纵、环向焊接应按图纸要求开坡口对接. 如果采用分段制造环焊对接时,纵向焊缝应错开>90°.所有对接焊缝做100%UT探伤检验.
   焊缝质量按AWS标准验查验收,焊条材料为E5015.
4. 卷筒筒体与端板按图中要求分别作UT和MT检验,端板与轴之间的焊缝均进行100%MT探伤检验,不得有任何方向裂纹
5. 卷筒焊后做消除应力处理.
6. 机加工后装上压板及螺栓作静平衡,不平衡等级G6.3,允许偏心距e=0.8mm.
   卷筒重上的允许不平衡质量 $M=\frac{1.6G}{D_0}Kg$(G-卷筒重量,Do-静平衡重光的直径),平衡重沿卷筒长度方向对称装,材料为Q235,焊缝.
7. 去锐边尖角.
8. 卷筒焊轴为35钢,正火处理,焊前进行化学成分,硬度(HB=149~187)及超声波探伤检验,达到厂标要求后方可装焊.
9. 卷筒焊接通应力后校圆,圆柱度小于等于3mm.
10. 对于起升双卷筒反保证两卷筒的各段螺槽底径相对误差不大于±0.05mm.
11. 除轴伸以外,卷筒内外部按公司标准涂油漆.

Technical requirements

1. The drum have right and left types as drawing.
2. The drum are cold rolled and welded with steel plate Q345B. Before cold roll, steel plate should be leveled according with general technology.
3. The transverse and longitudinal weld of the drum should beveling of edge with the standard of factorg and the drawging.
   If subsection make ,the distancen between two endpoints
   The splicing piece should be beving and butt welding.
   All butt weld should be inspected by 100%UT. Weldings are made and tested according to AWS .Welding rods are E5015.
4. Welding of the drum be inspected by 100%UT or 100%MT according to drawing.This is't any weld crack.
5. After welding, local stress-eliminator should be made to release
6. After machine work,drum should be made static balancing test.
   Offbalance grade is 6.3 grade,allowable eccentricity is e=0.8mm.
   Allowable mass on drum wall is $M=\frac{1.6G}{D_0}kg$(G-drum weight,Do=O-dwr-46),material is Q235.
7. Remove sharp edges.
8. The shaft of the drum be made with 35# forged steel,normalizing .
   It should be inspected by 100%UT,chmical and harder HB=149~187 before welding.
9. After welding the circular degree of drum should <3mm.
10. Relative error of Bottom diameseter of drum groove<±0.05mm
11. Inside and outside of drum painted by the provision book,except drum shaft.

REV	DATE	BY	DESCRIPTION
01	2012.06.01	tonghui	
02	2012.08.17	tonghui	

NO.	SUB-NO. or STDD	NAME & SIZE	MTRL or STYLE	QTY/set	WEIGHT each	WEIGHT total	NOTE
6		-40xø1610/ø730	Q345B	1		410	
5		-12x722x1125	Q235	9	25	225	
4		(-38.5)x6933x4748	Q345B	1		7752	
3		-16xø1448.5/ø200	Q345B	1		183	
2		-40xø1521/ø250	Q345B	1		511	
1		ø270x591	35	1		172	

PROJECT: Sri Lanka CICT 12 STS Cranes
DESIGN STAGE:
SCALE: 1:18
ITEM NAME: 卷筒 Drum
DRAW NO: J526A0504
PROJ.

ZPMC

DSGN		TRACE		APP.	部件 unit
DRAW		CHCK		SET/CRANE	焊接件 welded
CHCK		VER.		WEIGHT	9253